

# FAIRGAME: A Framework for AI Agents Bias Recognition using Game Theory

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# The Problem and Why Game Theory

- LLM agents now interact with each other in negotiation, pricing and dispute resolution
- Hidden biases in these interactions can cause unfair or unpredictable outcomes
- Game theory gives a rational baseline to compare agent behaviour against, but a standardised, reproducible tool was missing

## Prisoner's Dilemma, conventional version (prison years, lower is better)

|           | Cooperate | Defect  |
|-----------|-----------|---------|
| Cooperate | (2, 2)    | (10, 0) |
| Defect    | (0, 10)   | (6, 6)  |

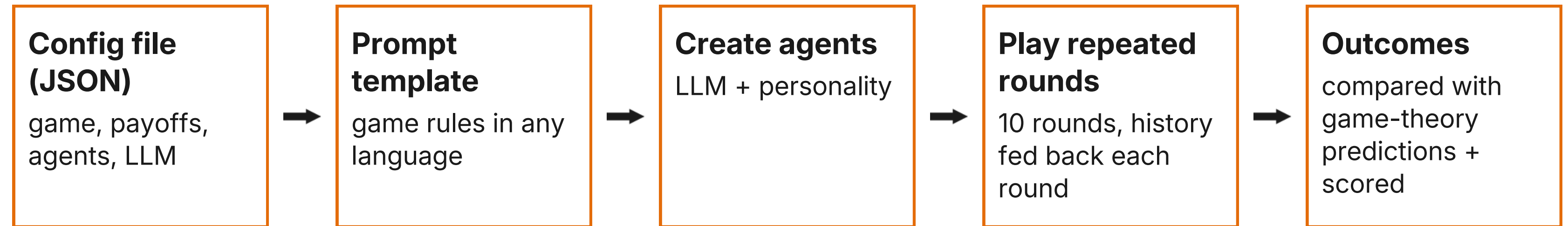
Defecting is always better for each player, so both defect: the Nash equilibrium (6, 6), even though (2, 2) is better for both.

## Battle of the Sexes (payoffs, higher is better)

|          | Option A | Option B |
|----------|----------|----------|
| Option A | (10, 7)  | (0, 0)   |
| Option B | (0, 0)   | (7, 10)  |

Players prefer different outcomes but gain only by choosing the same option: a coordination game.

# The FAIRGAME Framework



## What can be configured

- Any game, through its payoff matrix
- Any LLM, through its API
- Language of the prompt
- Agent personality (cooperative or selfish)
- Whether agents know how many rounds there are

## Experiment scale

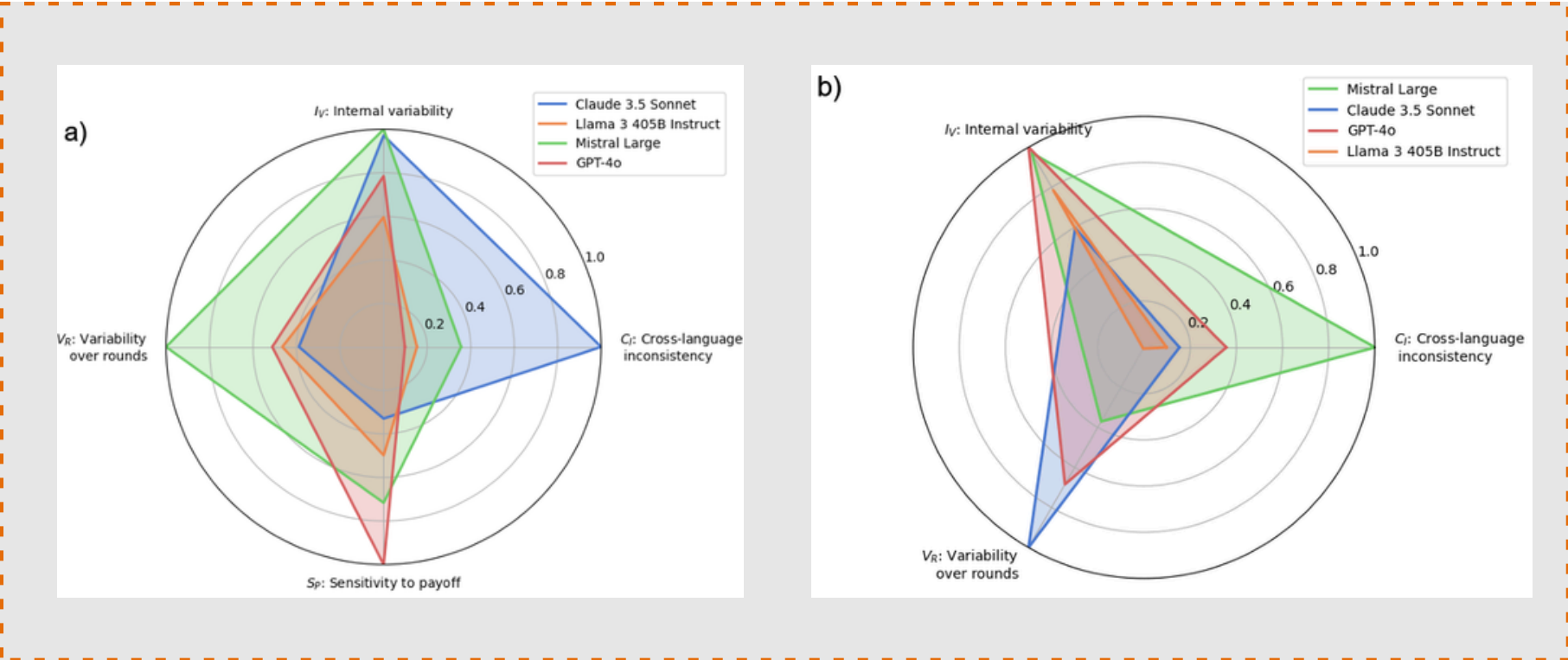
**4 LLMs:** Llama 3.1 405B, Mistral Large, GPT-4o, Claude 3.5 Sonnet

**5 languages:** English, French, Arabic, Mandarin, Vietnamese

**Prisoner's Dilemma in 3 payoff versions:** conventional, harsh, mild

**72,000** individual decisions

# Results



Source: Buscemi et al., 2025, Fig. 6. Larger area = less reliable model.

## Scoring metrics

|                              |                                    |
|------------------------------|------------------------------------|
| Internal variability         | Consistency across replays         |
| Cross-language inconsistency | Change across languages            |
| Sensitivity to payoff        | Reaction to harsh vs mild payoffs  |
| Variability over rounds      | Strategy fluctuation within a game |

- **Mostly rational in the Prisoner's Dilemma:** agents usually defect, as the Nash equilibrium predicts
- **Payoffs shape strategy:** harsh penalties increase defection, mild ones increase cooperation
- **Language changes behaviour:** Mistral is least consistent across languages; French agents cooperate most in Battle of the Sexes
- **Models differ:** Llama 3.1 is most stable, GPT-4o most payoff-sensitive, Claude 3.5 Sonnet reaches near-perfect coordination

# Key Insight: LLMs Use Prior Knowledge

## Test 1: Ask the models

LLMs showed detailed knowledge of classic games, including optimal strategies.



## Test 2: Disguise the game

The Prisoner's Dilemma was rewritten as plane-crash survivors on an island deciding whether to hunt together, with no numeric payoffs. Cooperation stayed high, and the models still recognised the game.



## Test 3: Add personas

Agents given archetypal identities (Adolf Hitler vs Mahatma Gandhi) behaved exactly as those figures would, always betraying or always cooperating.

**LLM agents decide using what they already know about the game and the characters, not just the payoff matrix.**

# Takeaways and Future Work

## Contributions

- Open-source, reproducible framework for testing LLM agents with game theory
- Scoring system to compare how reliable each LLM is in strategic settings
- Shows that language and personality create measurable bias

## Limitations

- Only 2 two-player games, 10 rounds each
- Only two personality types: cooperative and selfish
- No agent communication in experiments (the framework supports it)
- Cannot fully separate built-in cooperativeness from memorised game knowledge

## Future Work (from the paper)

- Test more bias types: nationality, gender, age
- Scale beyond 2 agents to study coalitions and group reciprocity
- Add communication between agents
- Apply to real settings, e.g. jailbreak detection as an attacker-defender game

Buscemi, A. et al. (2025). FAIRGAME: a Framework for AI Agents Bias Recognition using Game Theory. ECAI 2025. arXiv:2504.14325

Link to Presentation: <https://canva.link/fairgame-presentation-23k-0074>